MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP(Honours)/MGU-BBA(Honours)/MGU-BCA(Honours)Special Examinations, July 2026 (for 2024 Admission Regular Examinations, October 2025)

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCECO201 - QUANTITATIVE ECONOMICS - I

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Answer Scheme

Part A – Answer all questions. Each question carries 1 mark.

1. b) ∅

2. a) 0.7

3. d) C=a+bY

4. b) MR=MC

5. b) Values controlling a model/equation

6. a) f′g+fg′

7. a) Qd=f(P)

8. c) Slope of tangent line

9. d) Square matrix

10. b) 12

Part B Answer any 10 questions. Each question carries 3 marks

11. A demand function shows the relationship between the quantity demanded of a commodity and its determinants, mainly its price.

General form: $Qd=f(P)$

where:

- Qd = Quantity demanded
- P = Price of the commodity

Example: When price increases, quantity demanded generally decreases. (3 marks)

12. $TR=80x-2x^2$ (inverted parabola). (3 marks)

13. A−B= (1.5 marks)

B-A = (1.5 marks)

14. Find the minimum value of $f(x)= x^2-4x+5$

$f'(x)=2x-4$

For minimum,

$2x-4=0$

$X=2$ (1.5 marks)

Minimum value:

$f(2)=2^2-4(2)+5=4-8+5=1$

Answer: Minimum value = 1 at x=2 (1.5 marks)

15. A random experiment is an experiment whose outcome cannot be predicted with certainty before it is performed. (1.5 marks)

Example: Tossing a coin or rolling a die. (1.5 marks)

16. Let

A=

Transpose:

A^T= (1.5 marrks)

Again transpose:

$(A^T)^T$= =A (1.5 marks)

17. (i) $dy/dx = - 2x /3y^2$ (1.5 marks)

(ii) $dy/dx = 5x^3/3y^2$ (1.5 marks)

18. $P(A \cap B)=P(A)P(B|A)=P(B)P(A|B)$.

19. Given

$2x+3y=5$
$-3x-2y=1$

Determinant

D= =-4+9= 5
$D_{x=}$ = -10-3 = -13
$D_{y=}$ = 2 +15=17

Answer:

$x=-13/5, y=17/5x$.

20. AB exists; BA does not exist. (Value liberally)

21. $f'(x)=10(2x+3)^4$.

22. M13=0; (1.5 marks)

C22=3. (1.5 marks)

23. (i)2 (1 mark)

(ii)5 (1 mark)

(iii)3. (1 mark)

24. {1,2,3,4,5,6}.

25. A×B={(3,p),(3,q),(4,p),(4,q)}; (1.5 marks)

B×A={(p,3),(p,4),(q,3),(q,4)}. (1.5 marks)

Part C Answer any 5 questions. Each question carries 6 marks

26. Evaluate limit

Let $x^3-27=(x-3)(x^2+3x+9)$.
lim=(x^2+3x+9) at x=3 = 9+9+9=27.
Answer: 27.

27. Equilibrium income

S=-50+0.2Y, I=80.
At equilibrium S=I.
-50+0.2Y=80
0.2Y=130
Y=130/0.2=650.
Answer: Y=650.

28. Determinant A = -1 (Give 1 mark)

Cofactor of A = (Give 1 mark)

Adjoint of A = (Give 2 marks)

A^{1}= (Give 1 marks)

29. Differentiate (Answer any 3 - each answer carries 2 mark each)

1) Product rule: $y=(x^2+3x)(x^3+2)$

$dy/dx=5x^4+12x^3+4x+6$

2) $y=x/x^2+1$

$dy/dx= 1-x^2/(x^2+1)^2$

3) $y=(3x-5)/(x+2)$

Quotient rule ⇒ $dy/dx=11/(x+2)^2$.

4) $y=(2x^2+1)^{-4}$

Chain rule ⇒ $-16x(2x^2+1)^{-5}$

5) $y=\log(x^2+4)$

$dy/dx=2x/(x^2+4)$.

6) $y=7e^{3x}$

$dy/dx=21e^{(3x)}$

30. Continuity

Left value at x=2: $2^2+1=5$.

Right value: 3(2)-1=5.

f(2)=5.

LHL=RHL=f(2).

Answer: Continuous at x=2.

31. Role of Probability in Economics

Probability is an important tool in economics because it helps economists, businesses, and governments make decisions under conditions of uncertainty. Its major roles include:

1. Decision Making Under Uncertainty
2. Demand Forecasting
3. Risk Analysis
4. Insurance
5. Market Research
6. Economic Forecasting
7. Policy Formulation
8. Financial Markets

32. adj(A)=

33. Derivatives

a) $y=x \quad -7x^3-7x+2 \Rightarrow dy/dx=6x \quad -21x^2-7$ (2 marks)

b) $y=x \quad -3x^4+ 2x^2-x+1 \Rightarrow 5x \quad - 12x^3+4x-1$ (2 marks)

c) $y=x \quad -4x \quad -5x^3-x+6 \Rightarrow 9x \quad -28x \quad +15x^2-1$ (2 marks)

34. Independent & dependent events

Independent: one event does not affect another (coin toss and die roll). (3 marks)

Dependent: one event affects another (drawing two cards without replacement). (3 marks)

35. Matrix operations

A+B= (3 marks)

AB=(3 marks)